Cerebrospinal Fluid Drain - Full Protocol, MEDVAMC

1) INDICATIONS FOR PROPHYLACTIC CSF DRAINAGE:

1. Endovascular or open vascular repair/reconstruction procedures involving thoracic and/or abdominal aorta

AND

2. One of more of the following:
   - coverage of the hypogastric artery
   - >20cm extent of thoracic aortic coverage
   - prior open and/or endovascular aortic surgeries

Contraindications to CSF Drain Placement:

- Coagulopathy (INR > 1.5) or ongoing anticoagulant medication use
- Thrombocytopenia (platelet count <100K/mm3) or ongoing or recent (<6 days) use of antiplatelet agents other than aspirin
- Meningitis
- Infection at drain insertion site
- Cranial imaging demonstrating intracranial lesion or non-communicating hydrocephalus
- Prior paraplegia

2) POST-OPERATIVE SCHEDULE (subject to modification based on clinical exam and patient-specific factors):

<u>WITH prophylactic CSF drain:</u>

- POD 0-1: Drain open, MAP >90mmHg, q1hr neurovasc checks, Start DVT ppx 6 hrs postop
- POD 2: Drain clamped, MAP >90mmHg, q1hr neurovasc checks
- POD 3: Drain clamped, slowly bring MAP to >65mmHg, q1hr neurovasc checks. Hold PM DVT ppx
- POD 4: Drain out, q1hr neurovasc checks, resume DVT prophylaxis 6 hrs after drain removed
- POD 5: Out of SICU to floor, PT/OT

<u>WITHOUT prophylactic CSF drain:</u>

- POD 0-2: MAP >90mmHg, q1 neurovasc checks, hold SQ heparin, patient out of bed POD1
- POD 3: Slowly bring MAP to >65mmHg, q1 neurovasc checks, resume SQ heparin
- POD 4: MAP > 65mmHg, liberalize neurovascular checks, continue SQ heparin

3) MONITORING

Monitoring for Anterior spinal cord ischemia (with-or-without a CSF drain):
****q1hr x 48-72 hours**

- Assess lower extremity motor function (hip flexion)

- Assess lower extremity sensory function (ability to feel painful stimulus)
- Proprioception and vibration are transmitted via the posterior columns and will therefore be intact in presence of anterior spinal cord ischemia.
- Pupillary abnormalities
- Level of consciousness

Monitor distal pulses (per surgical team).

- Remember that distal pulses/signals can be intact, and presence of distal pulses doesn't rule out anterior spinal cord ischemia. (Upper and lower extremity vascular exams should be performed)

4) DRAIN MANAGEMENT & PREVENTION OF SPINAL CORD ISCHEMIA

****Key Tenets:**

Maintain spinal cord perfusion pressure > 90 mmHg
SCPP = MAP – CSF-pressure (*or CVP: whichever is highest*)
Maintain adequate oxygen delivery (oxygenation and hemoglobin)

a. Keep MAP > 90 mmHg (consider pre-operative baseline BP)

This allows for a spinal cord perfusion pressure of 90 mmHg – 10 mmHg = 80 mmHg.
Remember that a normal MAP is usually about 95 mmHg (BP: 120/80)
If CVP is high, (higher than CSF pressure) then MAP has to be increased
Management: Consider vasopressors or even epinephrine (vasoplegia vs low EF or RV dysfunction)

b. Keep CSF pressure 10-15 mmHg.

Leveling the transducer:

- Place the transducer (stopcock) at same height as the anterior iliac crest. (*surrogate for thoracic/lumbar spine*)
- All transducers (Art ,CVP, etc.…) should be at same height as lumbar drain transducer.
- Do not continuously transduce lumbar drain pressure. (Beware of risk of accidental injections into CSF)

Set the pop-off pressure to 10 mmHg:

- Raise the pop-off to 10 mmHg (on the printed scale) above the tansducer/stopcock
- If CSF pressure is greater than 10 mmHg (~ 14 cmH_2O) CSF will overflow and drain into collecting chamber.
- Normal CSF pressure is about 10-15 mmHg in supine position (<18mmHg once awake and moving legs)
- Normal CSF production is about 20 to 30 ml/hr (0.2 to 0.7 ml/min)
- Total CSF production is about 600-700 ml per day.

Conditions for changing pop-off/overflow setting: (For further details about troubleshooting, see section 6)

- No CSF drainage
- CSF drainage of <10 or > 40mL in 4-hour period without achieving the target CSF pressure
- If CSF drainage is 40mL over 2 hours regardless of CSF pressure

Avoid the following:

- draining a specific hourly amount of CSF
- over drainage. This increases risk of herniation, tearing of bridging veins etc.
- antihypertensives except in extreme situations.

  Please note:
  - Correlate Art line pressure with NIBP prior to starting antihypertensive.
  - Consider Clevidipine rather than nicardipine because of its shorter half-life
  - Increase vigilance and monitoring if antihypertensive is started.

c. Maintain adequate oxygen delivery

Maintain cardiac output (inotropes if needed for CI>2.5L/min/BSA) and oxygen saturation >92%.

5) DRAIN CLAMPING

Drainage is stopped and CSF drain is clamped on POD2 if:

a. No evidence of spinal cord ischemia
b. Stable hemodynamics
c. Low risk for spinal cord ischemia based on anatomy and extent of coverage of segmental arteries (posterior intercostal arteries etc..)
d. Discussion between ICU attending and surgical services

General notes for clamp trials:

- Patients can sit in chair with drain clamped. We prefer for them not to walk with clamped drain.
- Maintain designated MAP goals based on postoperative period and clinical status (also innumerated in section 2 above)
  - POD2: Drain clamped, MAP >90mmHg, q1 neurovasc checks
  - POD3: Drain clamped, slowly bring MAP to >65mmHg, q1 neurovasc checks
- Clamp trials and progression are based on the absence of neurological changes and continuous discussions between the ICU and surgical teams
- Maintain strict q1hr neuro exams for spinal cord ischemia during clamp time
- Check platelet count and INR the morning of anticipated drain removal
- Strict 4-hour bedrest post drain removal to avoid persistent CSF leak (activities like sitting and coughing will promote high CSF pressure and possible persistent CSF leak)

6) TROUBLESHOOTING

a. No CSF drainage:

Differential diagnosis:

- CSF drain is blocked / kinked
- CSF pressure is very low (previously drained too much)
- CSF catheter became disconnected / snapped off
- CSF catheter inadvertently pulled out of CSF space / lumbar area.
- Stopcock is turned off to the patient preventing a patent system.

Management:

- Lower pop-off pressure to zero mmHg temporarily. If CSF is draining, then drain is patent but CSF pressure is low. Raise the pop-off pressure back up to 10 mmHg. Continue to check patency and drainage q1hr.
- Check integrity of drain / stopcocks / lumbar area dressing.

b. Excessive CSF drainage:

Definition of excessive CSF drainage:

1. CSF drainage of<10 or > 40ml in 4-hour period without achieving the required CSF pressure
2. If CSF drainage is 40 ml in 2 hours regardless of CSF pressure

In general, excessive CSF drainage is a risk of brain herniation or subdural hematoma/tearing of bridging veins!

Differential diagnosis:

- incorrect leveling of CSF drain
- incorrect setting of pop-off pressure (height of draining chamber too low in relation to the CSF transducer level)
- obstruction to CSF drainage or inadequate absorption of CSF
- CSF overproduction -- this is very rare and shouldn't be high on differential diagnosis (consider infection/inflammatory causes)

Management:

- Full neuro exam including pupillary exam
- Raise pop-off pressure by 5 mmHg to decrease the amount of drainage per hour, and ensure no more than 10mL CSF draining over next 30 minutes.
- Ensure adequate MAP (>90) while decreasing CSF drainage depending on recent drainage.

NEVER clamp the drain unless discussed with ICU attending.

(clamping is usually only done during patient transport or mobilization out of bed or in anticipation for drain removal after 24 hours.)

c. Bloody / serosanguinous CSF drainage

This is a serious sign that needs prompt attention!

Differential diagnosis:
- subarachnoid hemorrhage
- subdural hematoma

Management:
- Notify the ICU attending AND Vascular Surgery team.
- Correct coagulopathy.
- Increase frequency of neuro checks (including pupillary exam)
- Emergent head CT scan to rule out intracranial hemorrhage
- Clamp the drain while ensuring adequate MAP.
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d. Drain snapped off / connection broke

This is a risk for: 1) infection / meningitis, and 2) unrecognized over-drainage.

Notify ICU attending, vascular surgery team AND cardiac anesthesia attending on call.
- If promptly witnessed then re-sterilization, cutting off "contaminated" tip and reconnection might be possible.
- Unwitnessed / unclear time: consider drain removal or cutting out a significant portion of visible drain and re-connect if replacement of drain is not feasible.
- Consider sending CSF fluid for culture and gram stain from the stopcock distal to the connector if obvious contamination is present.

e. Soiled CSF catheter dressing (dressings are never to be removed)

-if CSF drain is draining clear CSF, and the dressing itself is saturated with serous fluid (ie CSF), then the bedside SICU nurse can reinforce the dressing
- if CSF drain is draining clear CSF, but the dressing is bloody (ie indicating bleeding from the insertion site), then contact on-call CT anesthesia team.
-if CSF drain is draining bloody CSF, see section 6c above

7) MANAGEMENT OF SPINAL CORD ISCHEMIA

Delayed spinal cord ischemia typically presents in the first 48 hours, however in some it has been seen as late as 4 weeks. The physical exam is not always classical: There will often be a combination of motor paralysis or weakness as well as sensory deficits. An asymmetrical neurological exam does not rule out spinal cord ischemia. MRI or CT scan should NEVER be prioritized above starting treatment for suspected spinal cord ischemia. (Consider placing new drain if previously removed or if current drain is not patent. However increasing MAP far outweighs the benefit of placing a new drain.)

a. Notify ICU attending and Vascular Surgery team immediately
b. Meticulous neurological exam
    - Rule out epidural or subdural hematoma (correct coags and platelets)
    - Rule out central stroke
    - Stat neuro consult
    - Consider stat CT scan or MRI if high suspicion for epidural or subdural hematoma
    - Do **NOT** prioritize CT scan over initiation of treatment for spinal cord ischemia
c. Increase MAP: > 95-100mmHg, but may need to be higher if no response (95 mmHg +/- 20 mmHg)
    - Norepi and/or vasopressin or even phenylephrine
    - Epinephrine to increase cardiac output if needed.
d. Carefully drain additional 10 to 20 ml of CSF by lowering the pop-off
    - Monitor for other neurological changes (pupils, CNS) during this time
    - Monitor for blood in the drain.
e. Optimize oxygen delivery:
    - Increase cardiac output.
    - Increase hemoglobin >10
    - Optimize oxygenation (SpO_2)
f. Consider single dose of Solumedrol

8) PERSISTENT CSF LEAK POST DRAIN REMOVAL
There are no known prevention strategies, but there are a number of management options available:
- Consider draining CSF at pop-off of 10 mmHg for an hour or two prior to drain removal.
- Strict bedrest for 4 hours post drain removal to keep lumbar CSF pressure low.
- Prolong bedrest in severe ongoing cases
- Avoid excessive coughing and straining post drain removal.
- Consider stitch (skin and subcutaneous tissue
- Consult neurosurgery service for possible blood patch.

9) INTRAOPERATIVE CSF AND HEMODYNAMIC MANAGEMENT:
Will review the following ICP goals and drainage plan during preoperative timeout:
- Spinal cord perfusion pressure (=MAP – CSFP [or CVP if higher]) should be maintained at >65mmHg
- CSF Pressure should be maintained 10-15 mmHg
- CSF can be drained up to 20mL/h
- Hemoglobin should be maintained >10 g/dL
- Upon aortic cross-clamp application or complete aortic coverage during endovascular repair, MAP should be raised to be > 90 mmHg

10) REMOVAL OF CSF DRAIN

- ICU/vascular surgery team to hold anticoagulation at appropriate time (see table below)
- Vascular surgery senior trainee or attending to notify on-call cardiac anesthesiology attending that drain is ready for removal along with last anticoagulant dose, and results of the day's coagulation studies (INR ≤1.5, platelet count >100,000)
- Strict bedrest for 4 hours post drain removal to keep lumbar CSF pressure low.
- Dressing checks q1hr after lumbar drain removal
- Resume anticoagulation per protocol below and discussion with vascular surgery team

11) **ANTICOAGULATION STRATEGY**:

Standard antithrombotic protocol following TAMBE with lumbar drain in place:

- ASA 81mg PO QD
- Heparin 5000U SC TID
- No other antithrombotics while lumbar drain is in place unless cleared with anesthesia and vascular surgery
- Following lumbar drain removal:
  - Plavix 75mg PO QD may be added

Updated: 6/30/2025

MEDICATION	**NOTES**	**HOLD PRIOR TO REMOVAL**	**MINIMUM DELAY TO RESTART AFTER REMOVAL**
Heparin 5000 U SC TID	Standard protocol	6 hours	Immediately
Aspirin 81mg PO QD	Standard protocol	No need to hold	Immediately
Enoxaparin 40mg SC QD	Not standard protocol—should not be started when on ASA	12 hours	4 hours
Enoxaparin 30 SC BID	Should not be started with lumbar drain in	*12 hours	4 hours
Enoxaparin 1mg/kg SC BID	Should not be started with lumbar drain in	*24 hours	4 hours
Plavix 75mg PO QD	Should not be started with lumbar drain in. Start after drain removal.	*Can be removed immediately if started < 1-2 days prior	Immediately (non-loading dose)
Plavix loading dose	Should not be started with lumbar drain in. Start after drain removal.	*Consult anesthesia	6 hours (loading dose)
Heparin IV infusion	Should not be started with lumbar drain in	*6 hours + normal coagulation status	1 hour
Apixaban	Should not be started with lumbar drain in	*72 hours	24 hours

*For inadvertent administration. Notify anesthesia upon discovery of inadvertent administration.

For all other antithrombotics, consult with anesthesia.